

GMaster methods trust

Masked pseudo- C_ℓ on GPU: agreement and wall-clock vs NaMaster

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Abstract

We document *methods trust* for GMaster, a JAX/GPU reimplementaion of the NaMaster masked pseudo- C_ℓ (MASTER) estimator. The question is verifiable: on fixed maps, masks, and binning, does GMaster reproduce NaMaster’s decoupled bandpowers within numerical tolerance, and how do wall-clocks compare? We state the estimator equations, the map $\leftrightarrow$ alm pipeline, and a short note on agentic development and cross-checking. Two real-map demos at HEALPix $N_{\text{side}} = 4096$ —ACT DR6 temperature and a FLAMINGO Compton- y map—show fractional agreement at the 10^{-5} – 10^{-7} level and $\sim 9\times$ warm GPU speedups against a 192-core NaMaster reference. This note is *not* a cosmological analysis.

1 Verifiable problem

Partial-sky power spectra require propagating mask mode coupling. NaMaster [1] implements the MASTER decoupling of Alonso et al. [2] with a stable CPU reference (`pymaster`). GMaster exports a NaMaster-compatible API and runs the same algebra on accelerator arrays (JAX), targeting the full estimator chain rather than isolated transforms.

Trust is measured on *fixed inputs*: identical HEALPix maps and masks, linear bandpowers with the same $\ell_{\text{max}} = 3N_{\text{side}} - 1$, Jacobi mask refinement ($n_{\text{iter}} = 3$), and spin-0 temperature auto-spectra. We report

1. fractional bandpower agreement $C_\ell^{\text{GM}}/C_\ell^{\text{NM}} - 1$, and
2. wall time for field construction, coupling matrix, coupled cell, and decoupling (I/O excluded).

No cosmological parameters, foreground models, or survey-specific science enter this document.

2 MASTER estimator vs NaMaster

For fields a, b with masks M_a, M_b , pseudo-alm coefficients $\tilde{a}_{\ell m}$ are obtained by spherical harmonic analysis of $M_a a$. The *coupled* pseudo-spectrum is

$$\tilde{C}_\ell^{ab} = \frac{1}{2\ell+1} \sum_m \Re(\tilde{a}_{\ell m} \tilde{b}_{\ell m}^*), \quad (1)$$

with an extra factor of two for spin ≥ 1 as in NaMaster. Mask coupling mixes modes:

$$\langle \tilde{C}_\ell^{ab} \rangle = \sum_{\ell'} M_{\ell\ell'}^{ab} C_{\ell'}^{ab} + (\text{noise}). \quad (2)$$

Given workspace $M_{\ell\ell'}$, the decoupled bandpower estimate is

$$\hat{b}_j = \sum_{\ell \in \text{bin } j} W_{\ell j} (M^{-1} \tilde{C})_\ell, \quad (3)$$

with binning weights $W_{\ell j}$ from `NmtBin.from_lmax_linear`. GMaster’s `NmtField`, `NmtWorkspace.compute_coupling_`, `compute_coupled_cell`, and `decouple_cell` follow NaMaster’s call order and normalization; unit tests hold matrix elements and decoupled cells to NaMaster at float64 tolerance on synthetic skies.

3 Map↔alm pipeline

The timed chain mirrors NaMaster’s public workflow:

1. **Field.** Build `NmtField(mask, [map], n_iter=3)`; internally analyse the map and (lazily) the mask with spin-0 SHTs at $\ell_{\max} = 3N_{\text{side}} - 1$.
2. **Coupling.** Assemble $M_{\ell\ell'}$ for the field pair (scalar TT uses the optimized recurrence; polarised cases use Gauss–Legendre quadrature).
3. **Coupled cell.** Form $\tilde{C}_\ell$ from field alms (Equation 1).
4. **Decouple.** Apply M^{-1} and bin (Equation 3).

GMaster’s default scalar route at $N_{\text{side}} \geq 256$ uses a table-free latitudinal march (float32 recurrence with float64 accumulation) and optional float32 coupling operands where the v2 march is active; the demos below use this default production path, not a special “exact” override.

4 Agentic methods sketch

Implementation proceeded by (i) locking the NaMaster API surface and parity tests, (ii) replacing kernels incrementally (coupling, SHT, ring FFT) with JAX programs checked against NaMaster on CPU, and (iii) promoting only passes that improved wall-clock without breaking agreement gates. Large refactors were rejected when isolated benchmarks regressed accuracy or when device memory refused full- N_{side} geometries. Real-map overlays (this note) sit above the unit suite: they guard against footprint-specific failures that synthetic tests miss.

5 Trust evidence: ACT and FLAMINGO

5.1 ACT DR6 temperature

Map: ACT DR6 night PA4 f150 source-free coadd (Stokes I). Native HEALPix $N_{\text{side}} = 8192$; for this demo the map is degraded to $N_{\text{side}} = 4096$ with `healpy.ud_grade` and cached before estimation. Mask: finite, non-zero pixels ($f_{\text{sky}} \approx 0.48$), identical for both codes.

5.2 FLAMINGO Compton-y

Map: FLAMINGO L2p8 unlensed Compton-y lightcone, full sky at $N_{\text{side}} = 4096$ ($f_{\text{sky}} = 1$).

5.3 Results

Both cases use $n_{\text{lb}} = 50$, $n_{\text{iter}} = 3$, spin 0. Warm GMaster times exclude first-run JIT compilation; NaMaster used 192 CPU cores. NaMaster completed successfully in both demos—no “N/A” fallback was required at $N_{\text{side}} = 4096$ TT.

Table 1: Methods-trust summary from shipped `examples/*/README.txt` artifacts.

Case	$\text{rms}(C_\ell^{\text{GM}}/C_\ell^{\text{NM}} - 1)$	GMaster (warm)	NaMaster	ratio
ACT DR6 TT	1.4×10^{-5}	7.6 s	70.4 s	$\sim 9\times$
FLAMINGO y	2.3×10^{-7}	8.6 s	69.5 s	$\sim 8\times$

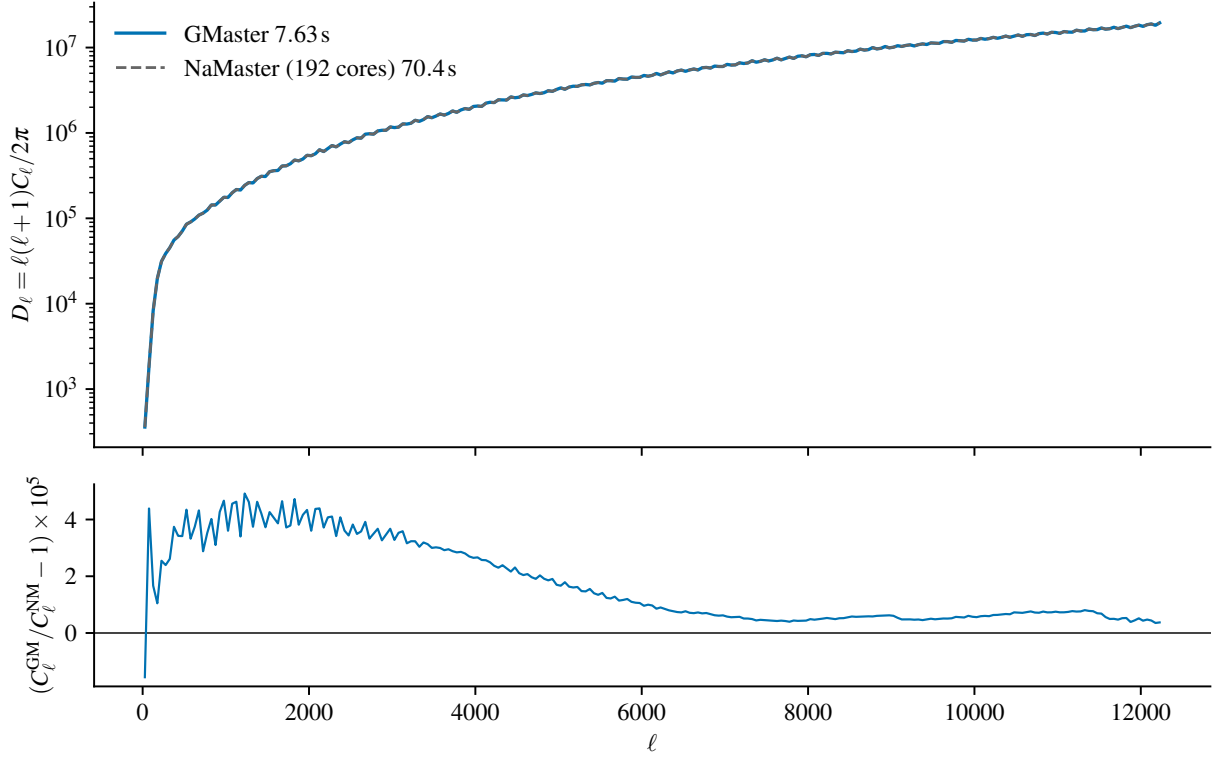


Figure 1: ACT DR6 TT: D_ℓ overlay (top) and $(C_\ell^{\text{GM}}/C_\ell^{\text{NM}} - 1)$ (bottom). Timings in legend from committed run logs.

6 Reproducing without re-running MASTER

Committed artifacts per case: `bins.npy`, `cl_gmaster.npy`, `cl_namaster.npy`, `overlay.pdf`, `README.txt`. The notebook `examples/namaster_vs_gmaster_timings.ipynb` reloads these arrays and rebuilds plots and Table 1; it does *not* invoke GMaster or NaMaster on the original FITS maps.

A Minimal API snippet

```
import jax
jax.config.update("jax_enable_x64", True)
import gmaster as gm

bins = gm.NmtBin.from_lmax_linear(lmax=3*nside - 1, nlb=50)
field = gm.NmtField(mask, [temperature_map], n_iter=3)
ws = gm.NmtWorkspace()
ws.compute_coupling_matrix(field, field, bins)
cl = ws.decouple_cell(gm.compute_coupled_cell(field, field))[0]
```

References

- [1] Alonso et al., *JCAP* **08** (2019) 017; `pymaster` 3.0.
- [2] Hivon et al., *ApJ* **567** (2002) 2.

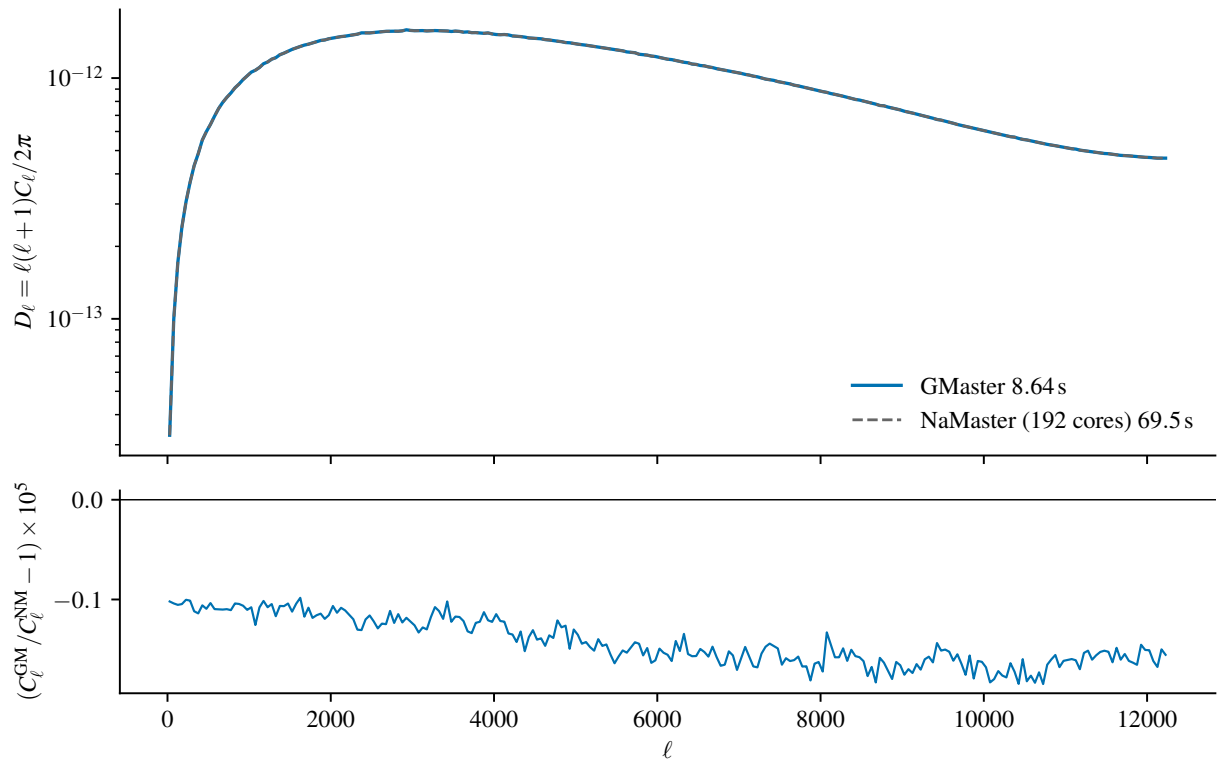


Figure 2: FLAMINGO L2p8 Compton-y TT: same layout as Figure 1.